



Questions Before Priors: Causal Dissonance, Insight, and the Origin of Strategic Theories

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Where do strategic theories come from?

- **The theory-based view:** economic actors are theorists — causal accounts of value creation direct what managers notice, test, and do (*Felin & Zenger 2017; Strategy Science 2024 special issue*)
- **A decade of progress on *holding* a theory:**
 - theory-guided experimentation and search (*Camuffo et al.; Chavda–Gans–Stern*)
 - theories as awareness advantages earning resources and rents (*Ehrig–Zenger*)
 - field evidence that theorizing pays (*Camuffo et al. 2020, and successors*)
- **But every formal model conditions on the theory being present** — as a menu to select from, a given conjecture, or a given awareness advantage
- The *origin* step is documented empirically (*Ott & Hannah 2024*) and named as the field's open analytical problem

Today: **a formal model of the step before the priors.**

A promise made in January 2025

Then — speculation (*the Utah talk*)

- Lonergan's *Insight*: experience → question → understanding → judgment → decision
- "Managers transcend pattern recognition when theorizing"
- The pre-Bayesian challenge: theory generation is a black box
- Closing slide: "*Formal linchpin: directed awareness — insight as pivot to new awareness states*"
- What was missing: **the mathematics**

Now — theorems

- Dissonance: the *detector* of a broken causal language
- Directed inquiry: what any answer must accomplish
- The price of a question — computable before its answers are conceivable
- Warrant: what the generating evidence does *not* license
- Equilibrium: when everyone lives in a Bayesian world

The structure of that talk survives intact. Today it has definitions, propositions, and proofs.

Why Bayesian machinery cannot be the answer

Two boundary facts:

- **No emergence.** Conditioning reweights a represented space; it cannot enlarge it. No sequence of updates makes an unrepresented causal distinction available for belief or action.
- **The posterior is self-sealing** (*Berk 1966*). When every represented theory is false, the posterior concentrates on the *least wrong* one — the agent grows **more confident**, not alarmed. Model failure is not an event in the agent's algebra.
- The economics of growing awareness (*Karni–Vierø; Schipper*) disciplines belief revision **after** expansion — but the expansion itself arrives exogenously
- So two objects are missing, and both are non-Bayesian:
 - a **detector** of representational failure
 - a **generator** of repairs — directed, costly, fallible

This paper: the full loop

dissonance $\longrightarrow$ inquiry $\longrightarrow$ refined awareness $\longrightarrow$ new models $\longrightarrow$ Bayesian judgment

- Experience **rejects every expressible theory** — dissonance, the detector Bayes lacks
- Rejection **localizes a question** without naming any answer
- A costly, fallible, truth-blind technology **may produce a new distinction**
- A bridge prior **rebuilds a probability space**; ordinary updating resumes
- All of it embedded in strategic interaction: **theories cause actions, actions cause evidence**

Back end: Kalai–Lehrer (1993) / Ryall (2003) — subjective rationality, self-confirming play.

This model *nests* them: hold awareness fixed and you recover the fixed-language world.

The new part is the front end: the language itself is endogenous.

Agenda

1. **A four-state laboratory** — two firms, four customer states; every object computed exactly
2. **The general model** — the same objects at full generality: mechanisms, tests, judged updating
3. **Key results** — pricing questions, the warrant theorem, epistemic absorption, persistence
4. **What it buys the theory-based view**



Part I

A four-state laboratory

The laboratory

- Two firms; unit market; product **architectures** $A_i = \{0, 1\}$
- Four **customer states**, arriving uniformly
- Three **causal mechanisms** the firms can conceive:
 - θ^0 : architecture 0 preferred
 - θ^1 : architecture 1 preferred
 - θ^n : architecture neutral
- Payoffs (winner-take-all market):
 - differentiate → preferred architecture wins: $(1, 0)$
 - pool on the preferred one → split: $(\frac{1}{2}, \frac{1}{2})$
 - pool on the wrong one → split β : $(\frac{\beta}{2}, \frac{\beta}{2})$
 - neutral state → $(\frac{1}{2}, \frac{1}{2})$ always
- Baseline today: $\beta = 1$ (perfectly forgiving demand)

	$c = 0$	$c = 1$
$r = 0$	s_{00}	s_{01}
$r = 1$	s_{10}	s_{11}

The objective truth — the *row law*:

$$\tau(s_{rc}) = \theta^r$$

Rows are causally relevant.

Columns are noise.

Nobody knows this.

Awareness: the language a firm can think in

- A firm's **awareness partition** collects the customer distinctions it can use — in theories, in beliefs, in policies
- An awareness cell is an implicit **causal claim**: "*these customers are alike*"
- **Expressible theories** assign one mechanism per cell
- Both firms start coarse — one cell, $\{S\}$ — so only three theories are thinkable:

$$M(\mathcal{P}^0) = \{m^0, m^1, m^n\} \quad (\text{constant assignments})$$

The key epistemic condition

The true row law is not among them. It is not *improbable* — it is **inexpressible**.

Unawareness $\neq$ probability zero: the firm cannot name it, bet on it, or condition a policy on it.

Private beliefs → differentiated commitments

- Firms hold private priors over $\{m^0, m^1, m^n\}$ — **heterogeneous**: firm 1 leans toward θ^0 , firm 2 toward θ^1
- Simple dominance calculation: architecture a is strictly dominant when

$$p^a > (2 - \beta) p^{a'} \quad (\text{at } \beta = 1: \text{ plain belief asymmetry})$$

- So the firms commit for the product generation: **firm 1 offers 0 everywhere, firm 2 offers 1 everywhere**
- Differentiated play is the market running an experiment **nobody designed**
- Neither firm can target "the informative state" — they cannot describe any state

The discovery generation

Episode	Actions (a_1, a_2)	Payoffs	Revealed
customer at s_{00}	(0, 1)	(1, 0)	architecture 0 wins here
customer at s_{11}	(0, 1)	(0, 1)	architecture 1 wins here

- m^0 fails the second episode; m^1 fails the first; m^n predicted $\frac{1}{2}$ at both
- **No expressible theory fits the record.** The active set is empty; Bayes' denominator is zero.

Causal dissonance

Every expressible theory is rejected. The question has a **determinate defect but no determinate answer**: *what difference between these customers explains the reversal?*

The posterior is not wrong. It is *gone*.

What if they had pooled?

- Same architecture at both states $\rightarrow$ each firm earns $\frac{1}{2}$ at every episode
- That outcome is consistent with **all three** constant theories $\rightarrow$ no dissonance, ever
- The coarse language **survives its own evidence indefinitely** — nothing marks it as broken

Exposure of a broken language has exactly two sources:

1. **Strategic variety** — rivals acting differently *inside your categories*
2. **Environmental harshness** — demand punishing uniform misservice observably

Hold that thought: at $\beta = 1$, source 2 is switched off. Everything today runs on source 1 — until the knife-edge slide.

Inquiry: directed, but not determined

What must any answer accomplish? A strict refinement of awareness whose cells are *coherent* — each cell admits a single mechanism consistent with the recoded record.

- Four states have **15** partitions
- Every restorative refinement must separate s_{00} from s_{11}
- Exactly **4** are minimally restorative:

Candidate	Cells	Complexity
rows	$\{s_{00}, s_{01}\}, \{s_{10}, s_{11}\}$	1 coordinate
columns	$\{s_{00}, s_{10}\}, \{s_{01}, s_{11}\}$	1 coordinate
isolate s_{00}	$\{s_{00}\}, \{\text{rest}\}$	2 coordinates
isolate s_{11}	$\{s_{11}\}, \{\text{rest}\}$	2 coordinates

- A factor-seeking (simplicity-filtered) technology produces **rows or columns** — and there, the evidence goes silent: $15 \rightarrow 4 \rightarrow 2 \rightarrow ?$

The row world and the column world

Define the **column world**: $\tau'(s_{rc}) = \theta^c$. It agrees with the row world at s_{00} and s_{11} — the two worlds generate **identical records**.

Proposition (Observational nonidentification)

No truth-blind inquiry rule formulates correctly in both worlds. A randomized truth-blind rule succeeds, worst case, with probability at most $\frac{1}{2}$.

Proposition (The price of robustness)

A language sufficient for *both* worlds is the full partition into singletons — every distinction the state space can carry.

Fallibility is the price of economy. An insight process that does not brute-force every distinction must be capable of being wrong. Whichever factor arrives, its unique consistent theory **extrapolates confidently to the two unlabeled states** — and in one of the two worlds, wrongly.

Pricing a question you cannot state

At dissonance the posterior is undefined and the answers are unrepresentable. What justifies paying γ for inquiry?

- **Leverage B :** count the retained episodes where the committed policy played the revealed *loser*; each is worth exactly $\frac{1}{2}$ per recurrence — **against every rival action**
 - rival repeats its winner $\rightarrow$ your switch converts a loss (0) into a split ($\frac{1}{2}$)
 - rival switches $\rightarrow$ your switch converts a split ($\frac{1}{2}$) into a win (1)

Participation rule

$B = \frac{1}{2} \cdot |\text{disagreeing episodes}|$. Elect inquiry iff $\lambda B \geq \gamma$ (λ : a *delabeled* success weight).

After discovery: $B_1 = B_2 = \frac{1}{2}$ — identical leverage from opposite experiences.

- **Answer-invariant:** the same value whichever of the four repairs arrives — the question is priced **without a menu of answers**
- The reasoning is delabeled: *"customers like the ones I lost will recur; whatever distinction inquiry yields would let me act on them"*

A question's value is action separation — not information

Record	Residual uncertainty	Leverage B
all episodes pooled	maximal — nothing pinned	0
discovery batch	nearly everything pinned	$\frac{1}{2}$

- **Rational cessation.** Run a validation generation: row-firm (correct) vs. column-firm (false). Policies differ off-diagonal; the row-firm wins both.
 - Winner's record agrees with its policy everywhere: $B_1 = 0$ — it **never inquires again**, and its awareness stays incomplete *optimally*
 - Loser's record disagrees twice: $B_2 = 1$ — dissonance, evidence, and incentive all land on the same firm

Sustained success extinguishes the question. Defeat concentrates it.

The natural inquirers in this model are the losers.

After insight: how much confidence is licensed?

Suppose the column refinement arrives. The recoded record pins a **unique** consistent theory — the new language awards it probability **one**.

Now assess reflectively: candidate worlds $\{\text{row}, \text{column}\}$, symmetric prior.

- Restored fit is **guaranteed by construction** — every restorative repair fits the data that generated it. Fit carries no information.

Theorem (The insight is not evidence)

With truth-blind production, conditioning on the insight's arrival adds nothing to the record. Warrant = $\frac{1}{2}$. Coherence = 1. **Warrant gap** = $\frac{1}{2}$.

- And the *same gap* afflicts the firm that happened to get it **right**: its coherence is also 1, its warrant also $\frac{1}{2}$
- **Unwarranted certainty is not a symptom of error.** Early confidence outruns evidence even when true.

The gap is invisible from inside

- Computing your own warrant gap requires *expressing both candidate worlds*
- By the robustness proposition, that takes the **full partition** — which no minimally-refined firm holds

So the overconfidence is invisible from within the representation that produced it. Warrant must come from outside:

- fresh evidence whose likelihoods differ across the worlds — one differentiated off-diagonal observation closes the gap completely
- an assessor with broader awareness (a board, an investor, a rival)
- a future self, after further refinement

Pooled observations — in any quantity — close **nothing**.

Warrant, like leverage, is purchased with differentiated action.

A market that believes its own mistake

Suppose **both** firms acquire the column refinement and become certain of the (false) column theory.

- Each plays the column policy → the firms **pool at every state** → every payoff is $\frac{1}{2}$
- $\frac{1}{2}$ is exactly what their shared theory predicts → **no dissonance, ever**
- No labeled episodes → $B_i = 0$ → even a *free, infallible* inquiry technology is declined
- The warrant gap sits at $\frac{1}{2}$ **forever**, under unbounded experience

- Subjectively optimal. Conjectures confirmed. Wrong at two of four states.
- This is a **self-confirming unawareness equilibrium** with payoff-relevant error — the market's shared false theory protects itself by eliminating the variety that would refute it

The knife edge: demand as epistemologist

Now let $\beta < 1$ — mis-served customers withhold $1 - \beta$ of the unit value.

- First generation containing an off-diagonal state: pooled firms earn $\frac{\beta}{2}$ where their theory predicts $\frac{1}{2}$
- The **payoff level itself reveals the label** — dissonance, leverage, and warrant closure arrive *together*

Knife edge

Payoff-relevant self-confirmation exists **iff** $\beta = 1$. With any outside option, consistency plus optimality force every recurring state to be correctly served.

- **The destroyed value is the corrective signal** — what demand withholds from the firms, their epistemics receives as evidence
- The two corrective forces, completed: **strategic variety** and **demand harshness**. Durable collective error requires the absence of *both*.



Part II

The general model

The general canvas

- n agents; finite states $s \sim \rho$; per-generation **committed policies**; consequences x_i = everything agent i observes; payoffs $u_i(x_i)$
- **Objective causal system**: elementary mechanisms $\theta : A \rightarrow \Delta(X)$; assignment $\tau : S \rightarrow \Theta$;

$$F(\cdot \mid s, a) = \tau(s)(a)$$

- **Actions cause consequences.** F is defined at every profile, played or not — that is the interventional content. No SCM formalism; micro-models (like the laboratory's customer market) generate mechanisms in examples.
- **Theories = one mechanism per awareness cell.** Why not arbitrary distributions? A saturated class can never be rejected — *coarseness alone cannot produce detectable failure*. A cell is a causal-homogeneity claim; dissonance is evidence of **mechanism heterogeneity inside a represented category**.
- Fixed awareness $\Rightarrow$ Kalai–Lehrer / Ryall (2003) exactly. The paper endogenizes their hypothesis space.

Dissonance becomes statistics

- **Consistency test:** theory m passes if empirical conditionals sit within tolerance η (plus vanishing slack) of m 's predictions at every visited (cell, profile) pair
- **Active set** = survivors. **Dissonance** = empty. The laboratory is the exact case $\eta = 0$.
- Recall the self-sealing posterior: under misspecification, Bayes gets *confident*, not alarmed
 - the test is therefore a **distinct faculty** — judgment, not understanding
- **Judged updating:** the posterior is restricted to survivors — belief never rides on a rejected theory

Exposure $\Leftrightarrow$ dissonance

Under a committed profile, an agent is eventually dissonant (a.s.) **iff** its awareness is exposed: every expressible theory fails somewhere on the visited path. Concealing profiles — the general form of the pooled trap — are exactly what keep false languages alive.

Direction, fallibility, and price — in general

Restoration is cellwise coherence

A refinement restores expressibility iff every cell admits a single mechanism consistent with its recoded record. Restorations must separate mutually incoherent evidence clusters; minimal restorations are typically **multiple**.

Nonidentification & the price of robustness

Worlds inducing identical record distributions receive identical outputs from any truth-blind technology — none is uniformly correct. Sufficiency for all candidate worlds requires the **join** of their causal partitions: economy of representation $\Rightarrow$ fallibility.

Participation (answer invariance)

Leverage B is computable from the record alone and **identical across every possible repair**. Elect iff $\lambda B \geq \gamma$. $B > 0$ only if some pinned distinction changes the robustly optimal action: questions whose every admissible answer supports current behavior are worthless at any positive cost — *however much model uncertainty remains*.

The insight is not evidence — in general

Theorem

With truth-blind production: the selection-adjusted warrant of a new formulation equals the ordinary reflective posterior **on the record alone**. The insight's arrival, and its retrospective success at organizing the experience that produced it, carry zero evidential weight. Among candidate worlds with identical record likelihoods, prior odds are preserved exactly.

Corollary (Self-audit)

Expressing the candidate set requires awareness sufficient for **all** candidates. An agent holding a minimal repair generally cannot formulate the comparison that measures its own warrant gap.

- The boundary, stated once: **everything evidential is in the record; nothing evidential is in the arrival of the insight**
- If production *were* correlated with truth (social transmission from an informed source), arrival would be evidence — truth-blindness is the polar case

When everyone lives in a Bayesian world

Definition (Self-confirming unawareness equilibrium)

(i) policies subjectively optimal; (ii) conjectures confirmed on path; **(iii) no dissonance — the language survives its own evidence; (iv) no priced question** — $\lambda_i B_i < \gamma_i$.

(i)–(ii) are classical. (iii)–(iv) are the new, *epistemic* conditions — and they carry the results.

Theorem (Epistemic absorption)

Almost surely: at most $n(|S| - 1)$ refinements; finitely many inquiry elections; and a finite generation after which **no language changes and no inquiry cost is paid** — every agent's epistemic life is Bayesian updating in a fixed language.

- The theorem is about the **death of inquiry**, not the convergence of play (classical, orthogonal, deliberately not claimed)
- One more rest state: **dissonant stasis** — the model is known-broken, but the break moves no decision: **rational anomaly tolerance**
- Grain-of-truth reading: refinement endogenously restores Kalai–Lehrer's condition *on the path of play*

Why performance differences persist

- Rest configurations can hold **false theories, heterogeneous languages, permanently different payoffs**
 - laboratory instance: row-firm earns $\frac{3}{4}$, coarse rival $\frac{1}{4}$ — forever, when the rival's question is priced below its cost
- Not because an equilibrium concept freezes them — because **the process that would erase them died economically**:
 - no dissonance (concealing play), or
 - no leverage (no action separation), or
 - the question priced below the cost of asking
- In **diagnostic environments** (the general $\beta < 1$), payoff-relevant error cannot survive at all

Strategic variety before probability: identical preferences, identical Bayesian competence, same objective world — different *questions* → different languages → different theories → different payoffs.

What this buys the theory-based view

- **Theory generation, formalized** — directed by defects, disciplined by restoration, fallible by necessity. The step every existing formal model assumes is now a model.
- **The value of a question is action separation, not entropy** — rational cessation for winners, concentrated incentive for losers
- **Founders' certainty outruns evidence even when correct** — organizing your own generating experience earns nothing; outside challenge is a *warrant technology*, not just monitoring. (The formal case for "test before you scale.")
- **Markets discipline theories through two channels** — captive demand + consensus $\Rightarrow$ durable collective error; the deviant entrant is the industry's *epistemic infrastructure*
- **Anomaly tolerance is an equilibrium phenomenon** — broken models persist rationally exactly where their breaks have no leverage

The research program

1. **This paper — questions before priors.** The origin of strategic theories: dissonance, inquiry, insight, warrant, absorption.
2. **From theory to plan.** Commitment under changing awareness: plans as *cached deliberation*; when confirmatory depth makes strategy rationally sticky.
3. **The theory-bearing firm.** How individual insights become shared languages, collective intentions, and an enduring center of agency.
4. **Competing theories and value capture.** Awareness rents, strategic disclosure, induced rival unawareness — joining this front end to the cooperative back end.

And beneath all of it, the deeper frontier: expanding the **mechanism vocabulary itself** — not just new distinctions, but new *kinds* of causes.

questions — before priors